Alex Ross
LinkedIn | GitHub | E-mail | Portfolio

Education

San Jose State University — Dec. 2025
Bachelor of Science in Software Engineering — *San Jose, CA*

Work Experience

Software Engineer — Jun. 2024 – May. 2026
Cinefind | Python, TypeScript, Next.js, React.js, Node.js, MongoDB, GitHub Actions, Docker — *Remote*

- Re-architected Cinefind into a production-grade platform, scaling to thousands of monthly users and securing funding and staffing support from LA-based accelerator programs
- Designed and implemented backend services to aggregate and normalize data from 20+ sources, delivering real-time email alerts for free movie screenings to paid subscribers
- Implemented one-click auto-RSVP from email alerts to automatically secure a user's spot in limited screenings
- Integrated Stripe webhooks for subscription payments, updates, cancellations, and automated user notifications
- Maintained CI/CD pipelines to automate testing, builds, and deployments on every merge to main

Projects

Spotter – Workout Companion
TypeScript, React Native, Expo, Node.js, PostgreSQL, Redis, OpenAI [App Store]

- Launched and maintained Spotter on the iOS App Store — a detail-oriented weightlifting app offering complete user control over workout structure, exercises, and performance analytics for personalized training
- Designed a 13-table schema with a three-level workout hierarchy served across 33 REST API endpoints
- Implemented native Apple and Google OAuth with cross-provider account linking for unified user profiles
- Introduced AI-powered exercise classification with Redis caching to assign muscle groups to user-created exercises
- Engineered autosave during active workouts via debounced API calls with minimal payloads to reduce latency

Duel — Real-Time Trivia
TypeScript, Node.js, Next.js, React.js, Redis, Docker, Nginx [Code]

- Built a horizontally scalable real-time trivia platform supporting concurrent multiplayer sessions via WebSockets, using Redis for distributed session state, matchmaking queues, and pub/sub messaging across servers
- Engineered low-latency game engine with turn timers, atomic Redis locking, reconnection grace periods
- Containerized the backend with Docker and Nginx, load balancing across instances with instance-pinned routing

Heat Check — AI Basketball Shot Tracker
Python, OpenCV, YOLO, TypeScript, React Native, Expo, Node.js, PostgreSQL [Code]

- Developed a mobile application that tracks basketball shots in real time using on-device computer vision and live camera input, enabling instant feedback during training sessions
- Trained a object detection model and engineered a shot classifier to detect shot outcome with 96% accuracy
- Converted model to TensorFlow Lite for on-device inference, optimizing for 30fps real-time performance

Twitter/X Bot Detector
TypeScript, Next.js, React.js, Python, Flask, Scikit-learn [Code]

- Engineered an end-to-end machine learning pipeline achieving 80% test accuracy on a 1M-user Twitter/X dataset, documented in Jupyter for team reproducibility and model transparency
- Curated raw user data, applied feature selection, and prepared inputs to maximize model training effectiveness
- Enabled model inference through an API and web app for live user lookup and classification with probabilities

Skills

Languages: JavaScript, TypeScript, Python, Java, Swift, C, SQL
Frontend: React.js, Next.js, React Native, Expo, HTML, CSS, Tailwind CSS
Backend: Node.js, Express.js, Flask, FastAPI, Spring Boot, GraphQL
Databases: PostgreSQL, MySQL, MongoDB, Redis
ML & Data: OpenCV, Scikit-learn, YOLO, Pandas, NumPy
Tools: Git, GitHub Actions, Docker, Nginx, AWS (EC2, S3, RDS), Selenium, Postman